1.1.4 Arbitrary Precision Calculations

When you use `//N` to get a numerical result, *Mathematica* does what a standard calculator would do: it gives you a result to a fixed number of significant figures. You can also tell *Mathematica* exactly how many significant figures to keep in a particular calculation. This allows you to get numerical results in *Mathematica* to any degree of precision.

expr`//N` or `N[`*expr*`]`	approximate numerical value of *expr*
`N[`*expr*`, `*n*`]`	numerical value of *expr* to *n* decimal digits

Numerical evaluation functions.

This gives the numerical value of π to a fixed number of significant digits. Typing `N[Pi]` is exactly equivalent to `Pi //N`.

```
In[1]:= N[Pi]
Out[1]= 3.14159
```

This gives π to 40-digit accuracy.

```
In[2]:= N[Pi, 40]
Out[2]= 3.1415926535897932384626433832795028841971
```

Here is $\sqrt{7}$ to 30-digit accuracy.

```
In[3]:= N[Sqrt[7], 30]
Out[3]= 2.64575131106459059050161575364
```

Doing any kind of numerical calculation can introduce small roundoff errors into your results. When you increase the numerical precision, these errors typically become correspondingly smaller. Making sure that you get the same answer when you increase numerical precision is often a good way to check your results.

The quantity $e^{\pi\sqrt{163}}$ turns out to be very close to an integer. To check that the result is not, in fact, an integer, you have to use sufficient numerical precision.

```
In[4]:= N[Exp[Pi Sqrt[163]], 50]
Out[4]= 262537412640768743.99999999999925007259719819
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.

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